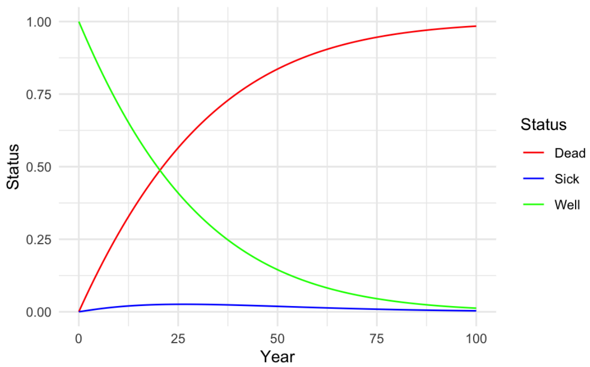
RDS 203
Tzu-Hsi (Cece) Jen
GitHub Repository: https://github.com/tzuhsi711/RDS203_ProblemSet2/tree/main
Deadline: April 30th, 2026 (11:59PM)

Problem Set 2

1. Markov Cohort Model

1a. The entire cohort begins in the Well state, while the proportions in the Sick and Dead states increase over time, with the Dead state increasing more rapidly (Fig. 1). As more individuals transition to the Sick and Dead states, the proportion in the Well state declines. Around year 20, the proportion in the Dead state exceeds that in both the Well and Sick states, indicating that more than half of the cohort has died, and the rate Dead state slows down as fewer individuals remain at risk. By the end of the cycle (year 100), nearly the entire cohort has transitioned to the Dead state. The median survival for this cohort is approximately 22 years, defined as the time at which 50% of the cohort has transitioned to the Dead state.

Fig. 1



1b. Similar to part 1a, the entire cohort begins in the Well state, while the proportions in the Sick and Dead states increase over time, with the Dead state increasing more rapidly overall (Fig. 2). However, the initial increase in the Dead state is slower than in part 1a due to lower age-specific background mortality at younger ages, and the rate of increase accelerates at older ages (age ≥ 60 years) as mortality rises. The Sick state exhibits a unimodal pattern, increasing initially as individuals transition from Well to Sick and then decreasing as recovery (Sick to Well) and mortality (Sick to Dead) outweigh new incident cases. The median survival for this cohort is approximately 66 years, defined as the time at which 50% of the cohort has transitioned to the Dead state. This reflects the importance of considering age-dependent mortality when modeling long-term survival outcomes.

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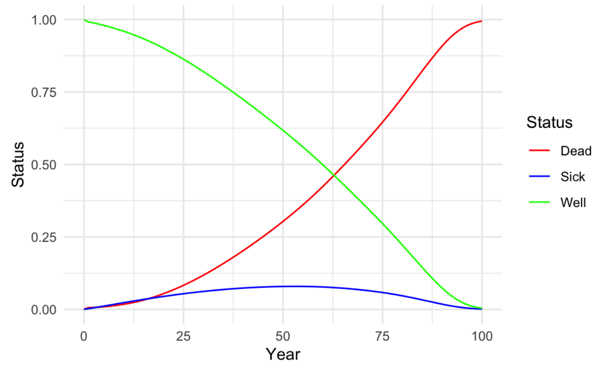
Fig. 2



1c. The lifetime risk of dying from the disease is approximately 48.76% while dying from other causes is approximately 50.73% for this cohort.

2. Monte Carlo Simulation

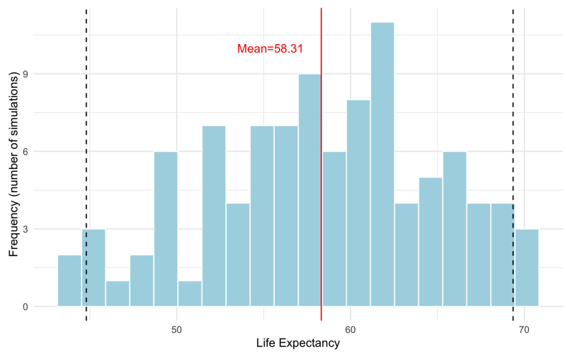
For the null hypothesis that cases occur uniformly at random within the city, the estimated p-value is 0.01917. As it is less than the 0.05 significance threshold, we reject the null hypothesis and conclude that the observed clustering of cases near the city center is unlikely to be due to chance.

3. Microsimulation

3c. The life expectancy for the cohort is approximately 57.82 years.

3e. The mean life expectancy is approximately 58.31 years, with a 95% uncertainty interval (UI) of [44.73, 69.51] across the 100 probabilistic sensitivity analysis (PSA) iterations. The distribution is moderately spread out, with most simulated values clustering around 55–65 years and some values in the lower and upper tails (Fig. 3). This reflects the wide 95% UI, suggesting that uncertainty in the model parameters leads to substantial variability in the estimated life expectancy.

Fig. 3



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4. Discrete Event Simulation

The ICU should have 36 beds to ensure a 95% probability that no patient is turned away over the next year.

5. Agent-Based Model

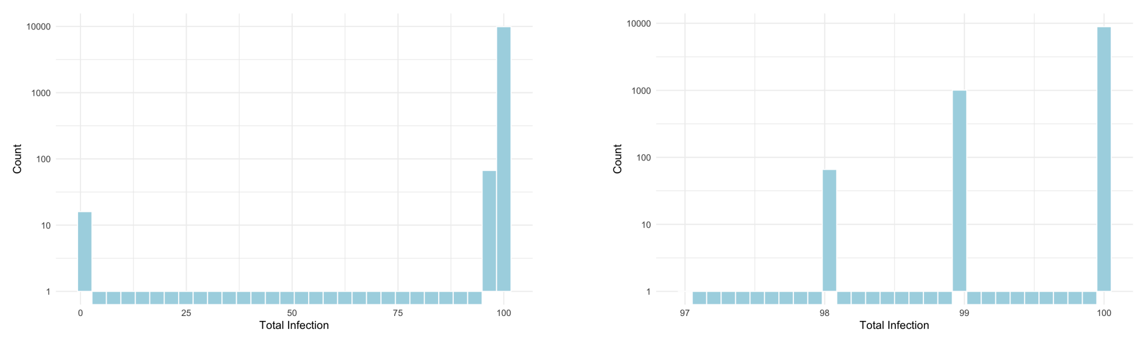
5e. The probability that the epidemic does not materialise is approximately 1.6%, indicating that around 98.4% of simulations result in at least one additional infection beyond the index case.

The distribution of the total number of infections was plotted for two scenarios: (i) all simulations, including those in which the epidemic did not spread beyond the index case (Fig. 4a), and (ii) simulations in which the epidemic spread beyond the index case (Fig. 4b). The counts (y-axis) in both scenarios were log-transformed to better visualise rare outcomes and improve the visibility of low-frequency events.

In scenario (i), a spike at 1 infection reflects simulations where the epidemic did not materialise (Fig. 4a). Aside from this, the distribution exhibits a left-skewed pattern with clustering near 100 infections. In scenario (ii), the distribution also exhibits a left-skewed pattern with clustering at 100 infections (Fig. 4b). This indicates that once initial transmission occurs, the epidemic tends to expand and infect nearly the entire population. This reflects the relatively high transmission probability and a fairly well-connected network, where sustained transmission enables the disease to spread widely across individuals.

Fig. 4

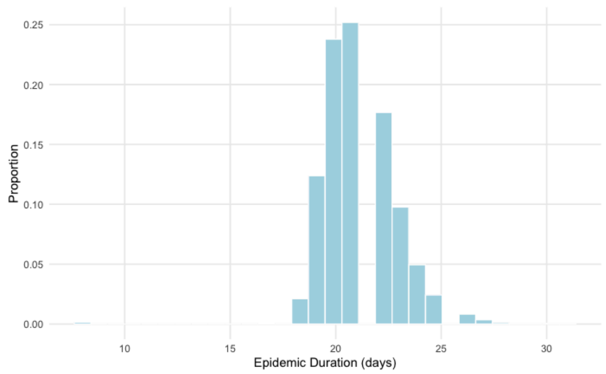
a. **b.**



The distribution of the epidemic duration is more spread out and approximately unimodal, with most simulations lasting around 20–23 days (Fig. 5). This suggests that although the total number of infections is often similar across simulations, the duration of the epidemic varies, likely reflecting a stochastic variation in infection patterns and transmission timing.

These results suggest that although there is a small probability (≈1.6%) that the epidemic does not materialise, once transmission is established, the outbreak tends to infect almost the entire population and persist for around 20 days.

Fig. 5

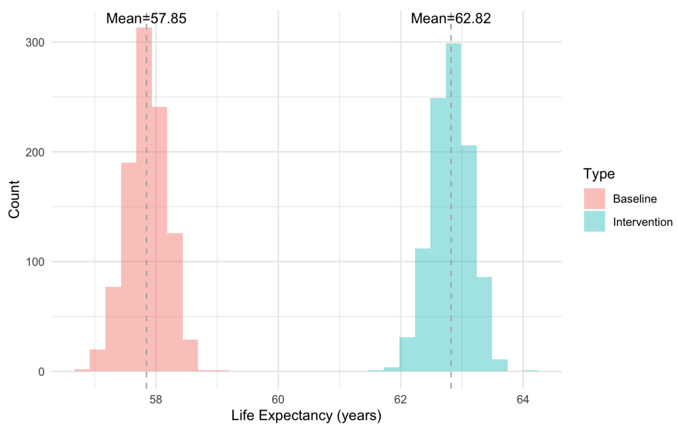


6. Common Random Numbers

6a. The average run time per iteration of simulation was 2.83 ms.

6b. The average run time per iteration was 3.21 ms. The mean gain in life expectancy was approximately 4.97 years, with a 95% uncertainty interval (UI) of 4.50 to 5.49 years (Fig. X). Following the intervention, the mean life expectancy increased from around 57.85 years to 62.82 years (Fig. 6).

Fig. 6



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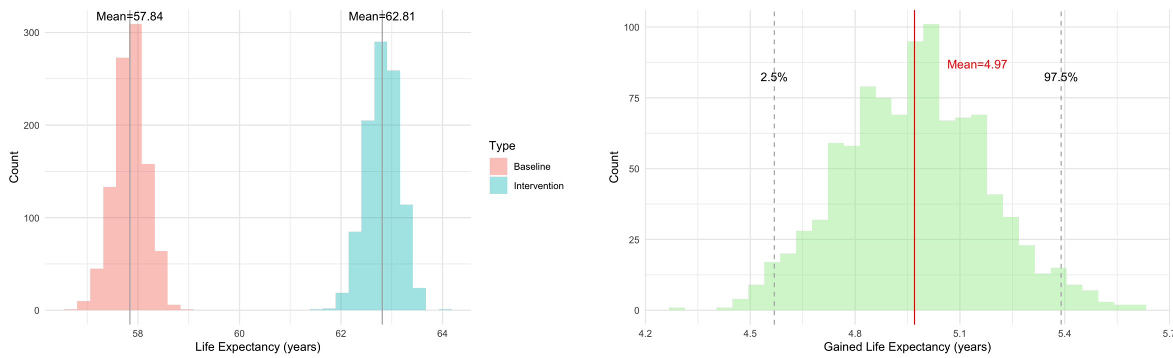
6c. The average run time per iteration was 3.76 ms. The mean gain was approximately 4.97 years, with a 95% UI of [4.57, 5.40]. Similar to 6b, the intervention increased life expectancy by around 4.97 years (Fig. 7a), but the 95% UI became narrower (Fig. 7b). Using common random numbers resulted in a slightly longer run time and a narrower 95% UI, indicating reduced variability in the estimated gain.

Fig. 7

a. **b.**

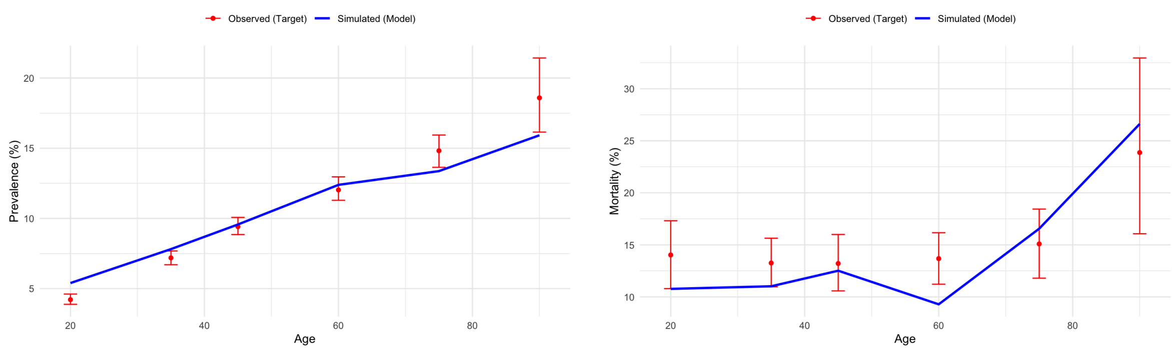


7. Microsimulation Calibration

7e. Calibration of the microsimulation model from Problem 3 to the observed data using greedy simulated annealing indicates a good fit. For both disease prevalence (Fig. 8a) and the 1-year risk of mortality (Fig. 8b), the simulated estimates from the best parameter set closely align with the observed values. The simulated values generally fall within the 95% confidence intervals (CI) of the observed data.

Fig. 8

a. **b.**



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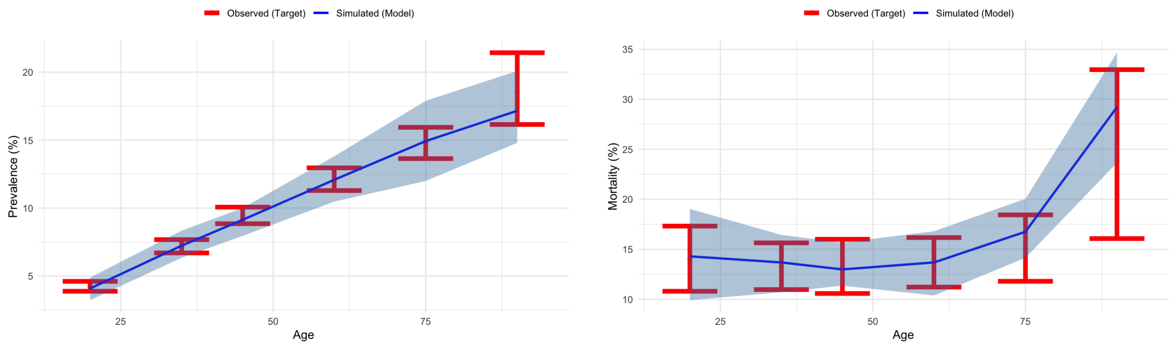
7f. The calibration was repeated 10 times to fit 10 different parameter sets. For both disease prevalence (Fig. 9a) and the 1-year risk of mortality (Fig. 9b), the simulated estimates closely align with the observed values. The simulated 95% UI generally falls within the 95% CI of the observed data. At older ages (age ≥ 60), the simulated UIs appear slightly wider than the observed CIs, indicating greater variability in the model estimates in this range.

Fig. 9

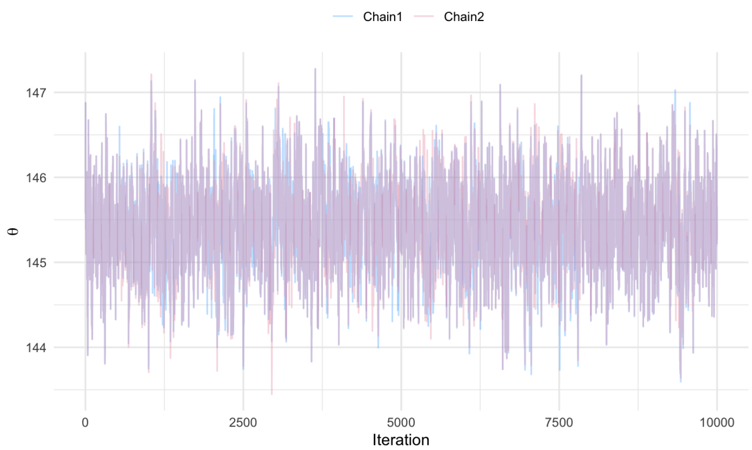
a. **b.**



8. Markov Chain Monte Carlo

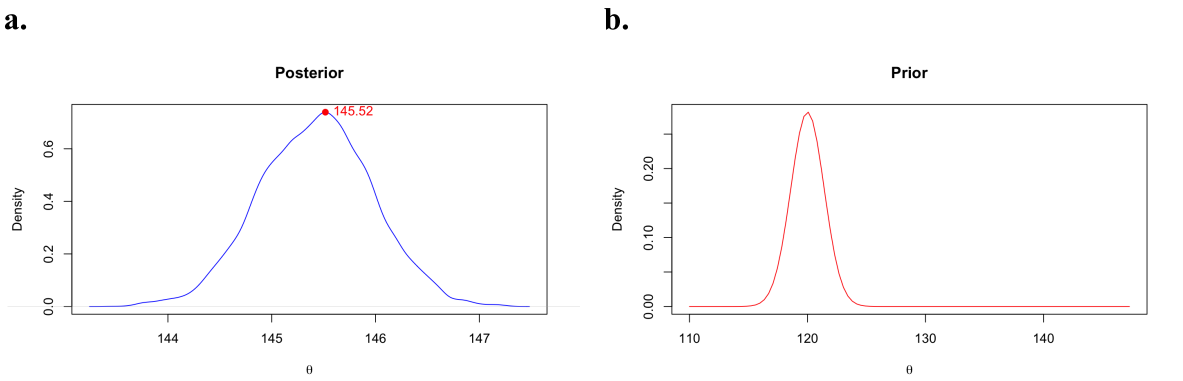
The Gelman–Rubin statistic is approximately 1.00 (0.99998), with an average acceptance rate of 74.81%, indicating good convergence between the two chains. The trace plots further support this, showing well-mixed behaviour with fluctuations around a stable region and no visible trends, suggesting that both chains are exploring the same parameter space (Fig. 9).

Fig. 9



Compared to the prior distribution, the posterior is shifted to the right and centred around 145–146, with a peak value (145.52) (Fig. 10a) higher than the prior mean (120) (Fig. 10b). This indicates a substantial update from the prior belief, implying that the posterior is primarily driven by the observed data rather than the prior.

Fig. 10



9. Approximate Bayesian Computation

Using an Exact Bayesian Computation (Exact ABC) rejection sampling approach, we obtained 0 accepted samples, resulting in an acceptance rate of 0.0%. This indicates that the probability of observing exactly 42 infections under the stochastic model is extremely low. As no posterior samples were obtained, only the prior distribution is shown (Fig. 11).

In our agent-based model from Problem 5, the epidemic size rarely takes intermediate values such as 42 infections, but instead tends to cluster around either 1 infection (no spread) or near 100 infections (widespread outbreak). Therefore, the probability of observing 42 infections under this model is very low, resulting in zero accepted samples under Exact ABC. This suggests that the current model may not adequately capture the observed epidemic dynamics. Additionally, this reflects a limitation of Exact ABC, where the inherent stochasticity of the model makes exact matching challenging.

Fig. 11